



Validating automated assessments of teaching effectiveness using multimodal data

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ABSTRACT

Background: For enhancing student learning in classrooms, high-quality teaching is essential. Research highlighted core dimensions of effective teaching, including classroom management, student support, and cognitive activation. However, traditional methods of assessing teaching effectiveness dimensions (e.g., student surveys) have limitations, including rating biases and resource intensiveness.

Aims: To overcome these challenges, we explored machine learning (ML) algorithms for the automated assessment of teaching effectiveness.

Sample: The study analyzed multimodal data—such as video, audio, and transcripts—from the Global Teaching Insights study, which included video recordings and transcripts from 46 teachers and 1,132 students in Germany.

Method: Scores for 18 teaching effectiveness subdimensions from three core dimensions were automatically generated by training attention-based ML models on multimodal features extracted from pretrained encoders. These ML-generated scores were compared with scores provided by human experts. A content validity study was conducted, where human experts evaluated the plausibility of ML-generated scores against human-generated scores. Structural equation models were used to assess the relationship between teaching effectiveness subdimensions and students' tested achievement.

Results: ML-generated scores were more reliable for some subdimensions (e.g., nature of discourse), and they were also plausible and content valid. ML-generated scores achieved higher absolute accuracy than human scores in 11 of 18 subdimensions. Limitations include reliance on human ratings as ground truth and inconsistent predictive validity, underscoring the need for refined models to generate actionable insights, such as real-time feedback systems.

Conclusions: The findings provide valuable insights for the development of automated feedback, enhancing the practical application of teaching effectiveness assessments.

1. Introduction

High-quality teaching is crucial for improving student learning in the classroom. Research has identified key dimensions¹ relevant to teaching effectiveness for both cognitive and motivational student learning outcomes (Pianta & Hamre, 2009; Praetorius et al., 2018; Seidel &

Shavelson, 2007), such as students' cognitive engagement and effective classroom management. Assessing and analyzing effective teaching and learning in real-life situations, such as school classrooms (i.e., ecologically valid settings involving natural, unstructured interactions; Blikstein & Worsley, 2016), is complex. Traditional methods face challenges, with interrater reliability often below .7 in observations

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¹ The terms *domain* and *dimension* are used variably across the field to indicate the highest level of aggregation when conceptualizing teaching effectiveness. We use *dimension* to refer to the highest level of aggregation and *subdimension* to indicate the level below dimension.

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(Bell et al., 2019) and high costs limiting scalability.

Traditionally, dimensions of teaching effectiveness are assessed with student ratings (Jaekel et al., 2020), teacher ratings (Wagner et al., 2016), or lesson observations (Praetorius & Charalambous, 2018). Still, these approaches bring challenges, including reliability, validity, and cost-effectiveness issues. Thus, it seems promising to investigate the potential to employ machine learning (ML) algorithms for an automated and alternative assessment of teaching effectiveness. Unlike traditional methods, ML can analyze teaching and learning interactions from a third-party perspective, integrating diverse classroom data sources such as lesson plans, board notes, and student work to enhance reliability and validity. Additionally, ML is cost-effective, potentially overcoming the limitations of manual classroom observations while bringing unique affordances and constraints. A decisive advantage of ML-based automated assessments of teaching effectiveness is that multimodal data can be analyzed efficiently, providing comprehensive and immediate (or even real-time) insights into complex teaching and learning interactions, extending prior work on single subdimensions (Hou et al., 2024) to a holistic evaluation of 18 subdimensions across three core dimensions.

A recent study showed that encouragement and warmth (a subdimension of student support) can be reliably assessed with ML-based methods applied to multimodal data, including video, audio, and transcript data (Hou et al., 2024). Moreover, in a follow-up study, the authors presented a multimodal ML model for automated classroom discourse assessment, integrating text, audio, and video to evaluate key discourse dimensions (Hou et al., 2025). The results showed that text played a dominant role, whereas acoustic features improved consistency with human ratings, achieving reliability comparable to human raters.

In this study, we extended the latter approach to additional subdimensions of teaching effectiveness to get a cost- and time-efficient but holistic (e.g., capturing the complexity of classroom dynamics comprehensively) understanding of complex teaching and learning interactions. This approach allowed us to compare automated assessments of key teaching effectiveness subdimensions by leveraging multimodal teaching data, including video recordings, audio data, and transcripts. We explored the extent to which dimensions of teaching effectiveness can be accurately predicted by automated methods compared to human expert raters and analyzed the reliability and validity of the automated scores.

2. Theoretical background

2.1. Teaching effectiveness as a crucial factor for successful student learning

Teaching effectiveness, encompassing classroom management, cognitive activation, and student support, has been demonstrated to be crucial in student learning processes and outcomes (Klieme et al., 2009; Seidel & Shavelson, 2007). For instance, classroom management strategies, such as establishing rules and routines, helped reduce disruptive student behavior (Oliver et al., 2011) and predicted student achievement (Korpershoek et al., 2016; Patall et al., 2023). The effects of teaching on student learning follow a relatively systematic pattern, with the strongest effects observed for domain-specific aspects of teaching that are most closely linked to students' learning processes (Seidel & Shavelson, 2007).

However, previous research has also shown that the effects are not always consistent, and the patterns of findings are somewhat more complex (Seidel & Shavelson, 2007). In fact, the effects of teaching effectiveness dimensions on student outcomes were confirmed in about half of the studies in a literature review on multilevel longitudinal designs by Praetorius et al. (2018). For instance, analyzing data from the 2019 Trends in Mathematics and Science Study (TIMSS) and its national extensions in Norway, Senden, Teig, and Nilsen (2023) found statistically significant positive relationships between classroom management and student achievement. By contrast, the effects of cognitive activation and student support varied, particularly across grade levels. Blömeke

and Olsen (2019) found strong positive relationships between teaching effectiveness and students' motivation across mathematics and science in students in Grades 4 and 8 across different countries. By contrast, the relationship between teaching effectiveness and student achievement varied depending on the academic subject, grade level, and country.

Against this backdrop, overly simplistic assumptions, such as direct linear relationships between teaching effectiveness and student learning, should be avoided (Vieluf & Klieme, 2023). Instead, high teaching quality should be understood as an opportunity to learn, which students may or may not utilize, meaning that whereas teaching effectiveness dimensions are necessary for successful student learning, they are not sufficient conditions. Klieme et al. (2009) proposed a theoretical model to explain the relationships between teaching effectiveness dimensions (e.g., classroom management) and student outcomes (e.g., achievement), in which teaching effectiveness is offered by teachers but students must actively engage with it (i.e., accept the offer). Consequently, the effects of teaching effectiveness dimensions on student outcomes unfold via student engagement processes, which act as mediators—specifically at the student level—such as time on task, depth of processing, motivation, emotions, and affect during learning (Alp Christ et al., 2024; Klieme et al., 2009).

2.2. Conceptualizations of teaching effectiveness

Conceptualizations of teaching effectiveness vary widely and include approaches that focus on teachers' effects on students' achievement (e.g., Rivkin et al., 2005), emphasize domain-specific teacher competencies (e.g., Kleickmann et al., 2013; Kunter et al., 2013), or define overarching dimensions of teaching effectiveness (Klieme et al., 2009; Pianta & Hamre, 2009). These dimensions are conceptualized as domain-general practices that transcend specific subjects, focusing on pedagogical interactions rather than content-specific methods. The present study followed the tradition of the latter approach, which focuses on overarching dimensions of teaching effectiveness that are crucial for successful learning and students' academic achievement. More specifically, this approach focuses on underlying quality subdimensions, such as the quality of the teacher's pedagogical interactions, meaningful cognitive engagement, support of students' academic and social development, and the clarity of instructional goals (Klieme et al., 2009; Lipowsky et al., 2009; Pianta & Hamre, 2009; Seidel & Shavelson, 2007) rather than on specific learning arrangements (e.g., teacher-led vs. student-led approaches) or specific methods (e.g., direct instruction, inquiry-based learning).

German-speaking countries commonly distinguish between three basic dimensions (TBD; i.e., *classroom management*, *student support*, and *cognitive activation*) of teaching effectiveness (Klieme et al., 2009; Praetorius et al., 2018, 2020), a differentiation that is also found in international work (e.g., Pianta & Hamre, 2009). According to the TBD framework, effective classroom management includes clear rules, routines, and proactive handling of disruptions, ensuring that learning time is maximized and students remain attentive and engaged. Student support emphasizes the importance of a learning environment that caters to individual student needs, including teacher sensitivity, fostering positive relationships between teachers and students, and providing emotional and social support. Finally, cognitive activation involves engaging students in intellectually stimulating tasks that promote deep understanding and critical thinking.

2.3. Assessing teaching effectiveness

The assessment of teaching effectiveness also differs between studies and countries (Klieme & Nilsen, 2022; Senden et al., 2022; Teig & Luoto, 2024), and any measurement of teaching effectiveness will inevitably have limitations (White et al., 2025). Teaching effectiveness is measured using a variety of approaches, ranging from *artifacts*, *teacher and student surveys*, *observations*, and *simulations* to the collection of *physiological*

data (Bell et al., in preparation). However, we focused on *student surveys*, *teacher surveys*, and *observations*, as they can capture the entirety of teaching in real classrooms.

Surveys are used in large-scale studies such as TIMSS and the Programme for International Student Assessment (PISA). Teachers and students hold different perceptions of lessons, resulting in divergence between their assessments of teaching effectiveness (Wisniewski et al., 2021).

Teacher surveys are valuable because teachers are essential stakeholders in a classroom. Surveying teachers offers insights that are challenging to obtain through other methods (e.g., teachers' internal state; Holzberger & Prestele, 2021). However, teacher ratings have low predictive validity and may be influenced by individual biases and classroom contexts, leading to variability in reliability and validity (Wagner et al., 2016).

Student surveys are valuable because students are also important stakeholders in a classroom and can provide valid insights into teaching effectiveness (De Jong & Westerhof, 2001; Fauth et al., 2014; Senden, Nilsen, & Teig, 2023). Moreover, they are time-efficient and cost-efficient, and predict students' achievement (cf. Jaekel et al., 2020, 2022). However, student ratings are idiosyncratic perceptions (Göllner et al., 2018) affected by rating tendencies (Göllner et al., 2016). Additionally, slight variations in item wording can influence the measurement of teaching effectiveness (Praetorius & Charalambous, 2018), and student ratings of teaching effectiveness show limited comparability across countries (Senden, Teig, & Nilsen, 2023).

Structured classroom observations by trained raters, often guided by standardized protocols, provide external and systematic evaluations of teaching practices (Praetorius & Charalambous, 2018). Classroom observations often have high content validity, and when conducted using recorded classroom videos, raters can take the time to review and reflect on classroom situations by rewatching and pausing the videos, allowing them to review artifacts and, for instance, zoom into the board, and also apply multiple teaching effectiveness frameworks. Well-known observation protocols, all of which share the assumption of the importance of underlying teaching effectiveness dimensions, include the Classroom Assessment Scoring System (CLASS; Pianta et al., 2008), the Unterrichtsfeedbackbogen (UFB, an observation system for school practice in German-speaking countries; Fauth et al., 2024), the protocol from the Global Teaching Insights (GTI) study (OECD, 2020a), and domain-specific protocols in, for example, mathematics (e.g., Mathematical Quality of Instruction [MQI]; Hill et al., 2008; Learning Mathematics for Teaching Project, 2011) or English (e.g., Protocol for Language Arts Teaching Observation [PLATO]; Grossman et al., 2014). For an overview of classroom observation manuals, see Bell et al. (2019), Klette and Blikstad-Balas (2018), and Praetorius and Charalambous (2018). However, classroom observations are resource-intensive in terms of time and financial cost, and are prone to unreliability (i.e., low interrater agreement; Bell et al., 2019; Göllner et al., 2016, 2018).

Due to the limitations of traditional approaches for assessing teaching effectiveness, it has been challenging to capture the complexity of classroom dynamics comprehensively. Thus, important research questions remain underexplored, such as the variability in teaching effectiveness dimensions within teachers over time (Polikoff, 2015), the dynamics of teaching effectiveness across multiple lessons by the same teacher (for a rare exception using traditional assessment methods, see Fauth, Wagner, et al., 2020), or the interplay between classroom compositions and teaching effectiveness.

2.4. Previous approaches for the automated assessment of teaching effectiveness

New technological advances, such as artificial intelligence (AI), including ML, offer great potential for gaining comprehensive insights into real classroom situations. For instance, multimodal approaches, such as assessments of audio, video, transcript, and physiological data

(Blikstein, 2013, pp. 102–106), can be used. Automatic AI-based assessments of teaching effectiveness dimensions can provide teachers with feedback on their teaching during or after lessons, enabling them to optimize their teaching practices (Grossman, 2018; Grossman et al., 2009; Kincade et al., 2020), such as using classroom management scripts (Wolff et al., 2021). At least two strands can be identified when focusing on AI-based applications in real classroom teaching.

As a first strand, there are numerous AI-based applications that support teachers' daily teaching activities (e.g., tracking students' progress and real-time performance, scoring written responses and essays). For instance, in a randomized controlled study with upper secondary English learners, Meyer et al. (2024) showed that GPT-3.5-turbo-generated feedback for essay revisions improved students' revision performance, motivation, and positive emotions compared with no feedback. The potential of Large Language Models (LLMs) to score student essays and thus relieve teachers was also shown in a recent study by Seßler et al. (2024), who evaluated the performance and reliability of different LLMs, including GPT-3.5, GPT-4, o1, LLaMA 3–70B, and Mixtral 8 × 7B for student essay scoring. Shi and Choi (2024) provided a systematic review of such AI-based applications in teaching.

As a second strand, several studies have focused on automatically capturing student behaviors (e.g., engagement and attention) and teacher behaviors (e.g., hand gestures, facial expressions, and body language) using behavioral data, often multimodal data, such as video and audio data (see Fütterer et al., 2025). For instance, Goldberg et al. (2021) explored the development and validation of automated methods for evaluating students' visible engagement during classroom instruction. Their research focused on creating a machine vision-based system that analyzes students' gaze, head pose, and facial expressions to determine their attention levels. Sümer et al. (2023) investigated automated methods for measuring student engagement during authentic classroom instruction by using audiovisual recordings, training deep embeddings—that is, compressed high-dimensional feature representations learned through deep neural networks—for attention and affective features, and developing engagement classifiers that achieve notable AUC scores. More examples include studies on the automated detection of students' hand raising (e.g., Bühler et al., 2023), engagement (e.g., Sharma et al., 2022), or mind wandering (e.g., Bosch & D'Mello, 2021; Bühler et al., 2024).

Besides these two strands, only a few previous research works have made initial attempts to automate the assessment of teaching subdimensions to gain insights into the quality of whole lessons. Some previous research has focused only on one modality (e.g., texts) to assess teaching practices. For instance, Liu and Cohen (2021) showed that automatically generated scores of interactive instruction, based on lesson transcripts, positively predicted value-added scores. A rare study using multimodal data by Pan et al. (2022) applied a multimodal ML framework to predict the quality of online teaching (i.e., clarity, classroom interaction, technical management of online teaching, empathy, and time management) and presented promising results regarding the capability of ML. Foster, Korban, et al. (2024) and Foster, Youngs, et al. (2024) evaluated the effectiveness of neural network models (i.e., BaS-Net, MLAD, and LSTR) for detecting teacher and student instructional activities in audio and video data from elementary classroom videos (i.e., a multimodal approach), focusing on low-inference instructional activities (e.g., students worked alone at a separate desk or in small groups, teacher moved around the classroom or not, teacher sat on a chair or not, student sat on floor or not, teacher nodded). Lazarides et al. (2024) employed a computer vision-based ML model to assess teachers' nonverbal immediacy (i.e., behavioral aspects such as gesture intensity, perceived physical distance, and facial expression) in classroom video data. Nonverbal immediacy mediated the relationship between teacher self-efficacy for student engagement and students' perception of enthusiastic teaching, ultimately influencing student interest.

The results of these studies demonstrate the potential for scaling

automated classroom observations. However, all models faced challenges in recognizing complex activities and relied heavily on high-quality data sets with observation scores for effective performance. Such approaches, which focus on teachers' visible behavior and have been dubbed *low-inference approaches*, are also used in commercial applications, such as *Speaker Coach* in PowerPoint from Microsoft. Speaker Coach is a tool that presenters can use to practice their verbal presentations by receiving personalized insights into their speech during a presentation (e.g., speaking pace, variance in speaking pace, use of filler words). However, such low-inference tools often overlook higher-order dimensions such as cognitive activation, limiting their utility for comprehensive assessments.

Another research group is exploring the automated detection of relevant teaching situations, working on the *Video Interaction Analysis Tool (VIAT)*, led by the National Education Lab AI (NOLAI) at Radboud University. This project aims to support secondary school teachers by enhancing their awareness and providing insights through reflection on their teaching experiences. The concept involves an AI-based system that collects metadata from relevant teaching situations, which teachers can use as a basis for reflection. The system is a work in progress. In the current development phase, the team is focusing on identifying the needs and requirements of teachers.

However, most existing automated approaches that assess teaching aspects tend to focus predominantly on specific, low-inference instructional features that are relevant but have limited ability to comprehensively cover students' learning activities and outcomes,² rather than on theoretically grounded and evidence-based dimensions of teaching effectiveness. A few exceptions focus, for instance, on automatically capturing classroom discourse (for asking questions, see Donnelly et al., 2017, pp. 218–227; Huang et al., 2020; for teaching discourse such as instructional speaking, see Jensen et al., 2020, pp. 1–13). For instance, Tran et al. (2023) presented promising results for natural language processing (NLP) techniques that automatically assess classroom discussion quality using rubric scores (e.g., the teacher presses students for information, and students support claims with evidence).

One of the few examples focusing on teaching effectiveness dimensions is the study by Wang and Demszky (2023), who utilized a ChatGPT model (gpt-3.5-turbo; zero-shot prompt) to predict observation scores based on the CLASS and MQI rubrics. They found that ChatGPT performed poorly at scoring teaching transcripts. For instance, the models consistently predicted the same scores for the same MQI or CLASS items, indicating that the predictions were more influenced by the dimension descriptions rather than the transcript evidence or prompting methodology. Additionally, the model tended to assign higher values to all CLASS dimensions than human ratings.

As another example, Bradford (2024) explored the usefulness of ML (e.g., logistic regression, random forests) to score teacher observations, focusing on aspects of project-based learning or teachers' integration of Next Generation Science Standards in teaching. She demonstrated that ML can improve scoring efficiency and accuracy relative to baseline random predictions. For instance, the baseline accuracy was 32 %, whereas ML methods achieved an accuracy range of 49 %–57 %. However, the results highlight ongoing challenges, such as overfitting and lower accuracy compared with human raters. Hou et al. (2024) demonstrated the potential of a multimodal approach for evaluating *Encouragement and Warmth*, a subdimension of student support, by combining facial and speech emotion recognition with large language models. This study leveraged the GTI data set, analyzing classroom videos with multimodal ML techniques and ChatGPT-based zero-shot annotations. The findings showed that these methods achieved

predictive accuracy comparable to human interrater reliability ($r = .51$), highlighting the importance of verbal and emotional cues in assessing this subdimension of teaching effectiveness.

In summary, with few exceptions, existing approaches to automating the assessment of teaching and learning interactions often focus on individual student activities rather than the entire classroom situation or on low-inference aspects, whose ability to comprehensively cover students' learning activities and outcomes may be limited.

2.5. Research questions

Despite progress in recent studies, a comprehensive study addressing the automated assessment of a broader range of teaching effectiveness subdimensions—spanning core dimensions such as classroom management, student support, and cognitive activation—remains lacking. Such research is crucial for advancing the automated, holistic evaluation of teaching effectiveness using established evidence-based criteria. AI systems that can process multimodal data sets (e.g., text, audio, and video) can enhance the assessment of teaching by gaining holistic insights (Lee et al., 2023).

Accordingly, we explored the possibilities of automated predictions of teaching effectiveness subdimensions using multimodal teaching data such as video recordings, audio data, and transcripts. We aimed to investigate whether it is feasible to imitate human ratings of teaching effectiveness subdimensions and thereby develop an AI-based rater, which would allow for the efficient generation of observational ratings of teaching effectiveness subdimensions without compromising reliability or validity. Regarding this objective, we investigated three research questions (RQs). Based on prior multimodal studies (e.g., Hou et al., 2025), we hypothesize that ML-generated scores will match human reliability in discourse-related subdimensions but may underperform in classroom management due to contextual nuances.

RQ1 addresses the reliability of ML-generated scores of teaching effectiveness subdimensions:

RQ1) How does the reliability of ML-generated scores for teaching effectiveness subdimensions compare to that of human-generated scores?

RQ2 and RQ3 address the validity of ML-generated scores of teaching effectiveness subdimensions. It remains crucial to investigate whether ML-generated scores of teaching effectiveness subdimensions can achieve content validity comparable to human-generated ratings, ensuring their reliability and applicability in educational research and practice.

RQ2) How does the content validity of ML-generated scores for teaching effectiveness subdimensions compare to that of human-generated scores?

As outlined in the theoretical background, teaching effectiveness is assumed to be important for successful student learning processes and outcomes. However, the strength of this relationship has varied (e.g., in some data, no effect or even negative effects were observed). For the specific GTI data we used, statistically significant relationships between the provision and use of core teaching effectiveness dimensions (e.g., effective classroom management) were not observed in the participating German classrooms (Grünkorn et al., 2020). Nevertheless, we aimed to explore the relationship between ML-generated scores and students' achievement to check whether ML-generated scores show the same patterns as human-generated scores, to gain insights into whether different variants of the ML-generated scores may help to improve the understanding of teaching effectiveness, and to generate additional hypotheses for future directions when elaborating more in depth on teaching and learning interactions.

RQ3) How does the predictive validity of ML-generated scores of teaching effectiveness compare to that of human-generated scores?

Importantly, we did not compare the teaching effectiveness scores of specific subdimensions derived from different sources (i.e., ML-generated, students, teachers, and observers), as these relationships are typically weak due to perspective specificity (Fauth, Göllner, et al.,

² We use the terms *dimension* and *subdimension* to refer to high-inference, theory-based instructional features, whereas *aspects* denote specific, low-inference elements that are relevant but capture only limited facets of student learning.

2020).

3. Method

3.1. Database, sample, and instruments

We used data from the GTI study (Klieme et al., 2023), which was part of the Teaching and Learning International Survey (TALIS) study conducted by the Organization for Economic Co-operation and Development (OECD) Teaching and Learning International Survey (TALIS) study conducted in Germany, aiming to better understand the quality of teaching practices (OECD, 2020a). The GTI study is alternatively referred to in the literature as the Talis Video Study (Ingram et al., 2020).

The data set we used comprised 92 video records of mathematics lessons focused on quadratic equations (40–90 min), data from 46 teachers ($M = 14.94$ years of teaching experience [$SD = 11.38$], 45 % female [binary variable—male/female]), and data from 1132 students ($M = 15.00$ years old [$SD = .81$], 53 % female [binary variable—male/female]) on the lower secondary level.

The same teaching unit was assessed in all participating classrooms. Two lessons (each lasting about 45–60 min) on the teaching unit (typically consisting of 8–12 lessons) were assessed for each teacher, depending on the school's schedule and curriculum. Data were gathered in the form of *video recordings*, *questionnaires*, and an *achievement test* that was administered before (t_1) and after (t_2) the teaching units (i.e., repeated-measures design).

3.2. Video recordings

Two cameras videotaped the lessons simultaneously at 25 frames per s. Camera 1 was positioned to focus on the teacher (i.e., recorded the teacher's movements, gestures, and interactions with the class), and Camera 2 was positioned to capture the entire classroom (i.e., recorded the layout of the classroom, student movements, and interactions among students). Multimodal data streams (i.e., audio, video, and transcripts) based on these recordings were available for analysis. However, it is important to note that the quality of the video data was constrained. For example, not all students were continuously visible throughout the recordings, which may limit the completeness of interaction analyses. Each video-recorded lesson was divided into 16-min segments (i.e., 367 16-min segments in total for the 92 video recordings), and two randomly assigned human raters scored each segment on 18 teaching effectiveness subdimensions of six components reflecting three core dimensions (see Table 1). The 18 teaching effectiveness subdimensions, the six components, and the three core dimensions were based on diverse perspectives from the participating countries and the existing research literature, but largely overlapped with the TBD (OECD, 2020a). For detailed documentation of the operationalization, see the Technical Report (OECD, 2020b). Two certified human raters scored each segment using the study's observation protocol (see the GTI's *Observation Tools*).

3.3. Questionnaires

Among other things, the questionnaires assessed students' and teachers' demographics (e.g., age, gender), perceptions of the teaching unit (e.g., disruption), and motivation (e.g., students' individual interest in mathematics, teachers' general feelings of self-efficacy). For an overview of all variables assessed in the GTI study, see the GTI user guide, codebook (OECD, 2020a, see also <https://www.oecd.org/en/about/projects/global-teaching-insights.html>), and scalebook (Herbert et al., 2021). We also provide an overview of descriptive statistics of items and scales used in this study via the OSF project.

3.4. Achievement test

We used *students' mathematics achievement* at posttest as the second dependent variable. The achievement test pertained to *quadratic equations*, a standard topic across educational systems worldwide (Grünkorn et al., 2020). The students were administered 30 (pretest) and 25 (posttest) items (e.g., multiple-choice, modeling problems) covering quadratic equations to assess procedural skills and basic understanding, transformations (e.g., completing squares) to assess conceptual knowledge, and deriving equations for real-world scenarios to assess problem-solving and application skills. All items were scored for the accuracy and completeness of the solution, and item response theory (IRT) scores (i.e., weighted likelihood estimates; Warm, 1989) were used to ensure a more precise and reliable measurement of student performance. For an example item from the mathematics test used in Germany, see Grünkorn et al. (2020). The pretest and posttest in our study demonstrated perfect internal consistency (KR 20).

3.5. Data analysis

3.5.1. Automated assessment of teaching effectiveness subdimensions

In Step 1, we predicted human-generated scores in the teaching effectiveness subdimensions (see Table 1) following the approach introduced by Hou et al. (2025). A detailed description of the pre-processing routines, model pipeline, and training setup can be found in the paper by Hou et al. (2025). Fig. 1 depicts our proposed multimodal architecture, which integrates rich information from diverse data streams (i.e., video, transcript, and audio) and leverages multitask learning to assess multiple dimensions of teaching practices simultaneously.

3.5.1.1. Unimodal feature extraction. First, we employed state-of-the-art pretrained models as unimodal encoders to process each modality independently and generate respective embeddings (i.e., feature vectors consisting of numerical values), serving as input for the subsequent fusion and prediction modules. These models are typically trained on vast amounts of data, equipping them with the capability to capture sophisticated semantic features from raw signals. For the text modality, we utilized JinaBERT-Base-de,³ an advanced text-embedding model capable of handling long German sequences up to 8,192 tokens, particularly suitable for our extensive transcripts. The model converts each word into a 768-dimensional embedding vector, which we subsequently aggregate at the utterance level to optimize computational efficiency. Similarly, we employed XLSR-German⁴ and clip-vit-large-patch14-336⁵ to extract acoustic and visual representations from the audio and video data for every 10-s chunk.

3.5.1.2. Multimodal fusion. Different modalities likely provide complementary information featuring diverse perspectives on classroom dynamics. Drawing inspiration from recent advances in multimodal affective computing (Shi & Huang, 2023, pp. 14752–14766; Wu et al., 2024, pp. 3588–3602), we adopted a text-driven multimodal fusion approach based on attention mechanisms (Vaswani et al., 2017). The mechanisms mirror human cognitive processes by selectively focusing on salient information while filtering peripheral details. As illustrated in Fig. 1, our architecture incorporated two sequential cross-attention encoders to facilitate intermodal interactions through text-guided retrieval of relevant features from the audio and video modalities. A subsequent self-attention encoder processed the temporal dynamics of the fused multimodal representations.

³ <https://huggingface.co/jinaai/jina-embeddings-v2-base-de>.

⁴ <https://huggingface.co/facebook/wav2vec2-large-xlsr-53-german>.

⁵ <https://huggingface.co/openai/clip-vit-large-patch14-336>.

Table 1

Conventionalization of teaching effectiveness in the global teaching insights study.

Core dimension	Component	Subdimension	Indicator
Classroom Management (i.e., “the ability to orchestrate smooth and efficient lessons”)		Routines (i.e., “Are routines used for the smooth handling of non-instructional tasks that happen in the classroom on a regular basis (e.g., marking attendance, making announcements, organising into different groups or passing study material)?”)	• Time on task • Activity structure and frequency • Time of lesson
		Monitoring (i.e., “Does the teacher monitor the classroom space and students by walking around the room or by scanning the classroom?”)	
Social-emotional support (i.e., “the ability to create a safe learning environment for all students”)		Disruptions (i.e., “Does the teacher take swift action to handle disruptions and breaks in the flow of lessons (e.g., equipment failures, outside interruption, misbehaviors)?”)	• Persistence • Requests for public sharing
		Respect (i.e., “Does the teaching build a mutual culture of respect for communication and interaction between students and with the teacher?”)	
Instruction	Discourse (i.e., “the ability to start and manage instructional dialogue with students”)	Encouragement and warmth (i.e., “Does the teaching build a well-considered and positive learning environment that includes sharing moments of warmth and encouragement?”)	• Discussion opportunities
		Risk-taking (i.e., “Does the teaching create an environment where students feel comfortable seeking guidance or sharing their thinking with everyone?”)	
		Nature of discourse (i.e., “To what extent do students have opportunities to direct, lead and engage in classroom talk clearly focused on a learning objective?”)	
		Questioning (i.e., “Does the teaching include questioning that engages students in deep thinking (e.g., analysis, abstraction, synthesis, justifications, extensions and conjectures)?”)	
		Explanations (i.e., “Does the teaching involve detailed explanations of ideas or procedures, which are substantial, coherent and focus on deep features of the topic?”)	
	Quality of subject matter (i.e., “the ability to handle content with depth, accuracy and coherence”)	Explicit connections (i.e., “Does the teaching succeed in making explicit, detailed and elaborate connections between subject matter ideas, procedures, perspectives, representations, and experiments?”)	• Explicit learning goals • Accuracy • Real-world connections • Connecting mathematical topics • Mathematical • Types of representation • Organization of procedural instruction • Metacognition • Repetitive use opportunities • Technology for understanding • Classroom technology • Student technology • Software use for learning
		Explicit patterns and generalizations (i.e., “Does the teaching identify strong patterns between different aspects and encourage students to generalise from specific examples to broader foundational concepts or definitions?”)	
		Clarity (i.e., “Does the teaching present ideas, concepts, tasks and teaching in a clear and coherent manner that is easy for the students to follow, with no instances of confusion left unaddressed?”)	
		Engagement in cognitively demanding subject matter (i.e., “Does the teaching place strong intellectual demands on the students by requiring them to engage in genuine analysis, creation, evaluation and thoughtfulness, through demonstration, discussion or practice?”)	
		Multiple approaches to/perspectives on reasoning (i.e., “Does the teaching create opportunities for students to use two or more procedures or reasoning approaches when working with specific content, questions or challenges?”)	
	Student cognitive engagement (i.e., “the ability to stimulate and challenge students’ thinking”)	Understanding of subject matter procedures and processes (i.e., “Do students have the opportunity to make sense of the rationale behind the individual steps of procedures, methods and explanations?”)	
		Eliciting student thinking (i.e., “To what extent does the teaching make use of questions, prompts, tasks or activities to make students not only share their answers but also present their reasoning of and thinking?”)	
		Teacher feedback (i.e., “Is the teaching responsive to student understanding and contributions through back and forth exchanges between students or with the teacher to give detailed feedback on student thinking and work?”)	
		Student-aligned instruction (i.e., “Is the teaching shaped by student contributions and aligned with student ideas so that student understanding and learning can be improved?”)	
		Assessment of and responses to student understanding (i.e., “the ability to uncover students’ thinking to guide further instruction”)	

Note. This overview is adapted from the OECD report (2020a) and the definitions are adopted from the OECD observation tool (OECD, 2020a).

3.5.1.3. Multitask learning. This study aimed to provide comprehensive insight into the teaching effectiveness assessment across six components, yielding 18 subdimensions (Table 1; OECD, 2020a). However, constructing an individual model for each subdimension would be resource-intensive. Thus, we employed a multitask learning approach

that enabled a model to assess multiple subdimensions simultaneously. Specifically, we trained six specialized models for the six components. Each model was responsible for generating predicted scores over the three corresponding subdimensions. This method optimizes computational efficiency and attempts to capture interrelations between

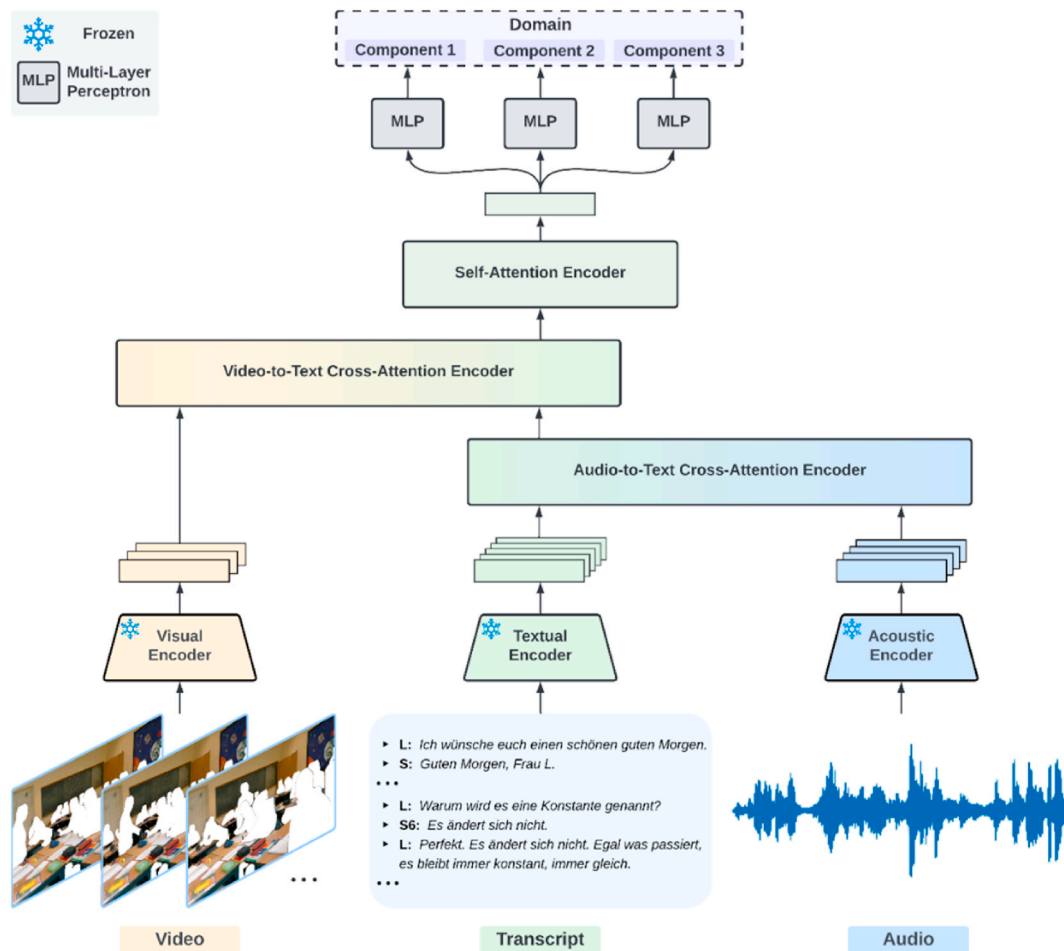


Fig. 1. Multimodal Architecture for Automated Assessment of Teaching Effectiveness
 Note. The architecture shown was adopted from Hou et al. (2025).

different subdimensions of teaching effectiveness. Furthermore, we applied a loss function called Ordinal Log-Loss (Castagnos et al., 2022, pp. 4604–4609) to address the ordinal nature of rating scores and class weights to tackle data imbalances (see Fig. 2).

3.5.2. Implementation

We applied a teacher-independent five-fold cross-validation approach to model construction. The data set (i.e., 367 segments) was divided into five equal parts, each serving as a test set once, while the remaining four parts were used for training. By rotating through all folds, we gathered predictions for each segment in the data set. Moreover, to evaluate a model's generalizability to new teachers, all segments from the same teacher were kept within the same fold. The choice of five folds reflects a balance between computational efficiency and evaluation stability, particularly given the limited number of teacher identities in the data set. The average human rating for each segment served as the ground truth for model training.

3.5.3. Reliability of ML-generated scores (RQ1)

In Step 2, to investigate the reliability of the ML-generated teaching effectiveness subdimension scores, we compared these scores with the arithmetic mean of the two human-generated scores (i.e., interrater reliability) utilizing three approaches inspired by research on the measurement of metacognitive monitoring (Schraw, 2009). We investigated *absolute accuracy* (i.e., the discrepancy between ML-generated and average human-generated scores; focusing on the correctness of individual scores), *relative accuracy* (i.e., the correlation between a set of ML-generated and human-generated scores; focusing on consistency),

and *bias* (i.e., the degree of underestimation and overestimation indicating the direction and magnitude of the error of ML-generated scores when considering the human-generated scores as the *truth*). A positive bias indicates that the ML-generated scores are lower than the human-generated scores (underestimation). By contrast, a negative bias indicates that the ML-generated scores are higher than the human-generated scores (overestimation). We descriptively compared the accuracy and bias with those observed for the two human-generated scores of two different raters. The terms absolute accuracy, relative accuracy, and bias index are rooted in metacognitive research and have been adopted here to quantify the agreement between human- and machine-generated scores. We note that these terms may be used differently across disciplines—for instance, in computer science, accuracy typically refers to the proportion of correct classifications in binary or multiclass settings. In contrast, our usage focuses on continuous or ordinal scores and follows conventions from psychological assessment research.

See Appendix A for the formulas for all three approaches. In addition, we have included confusion matrices for all models and all 18 subdimensions in our OSF project.

Analyses for all three RQs were performed using the statistical program R version 4.3.2 (R Core Team, 2023) and RStudio version 2023.12.0.369 (Posit team, 2023).

3.5.4. Content validity of ML-generated scores (RQ2)

We assessed human plausibility to gain insights into the content validity of the ML-generated scores. We randomly selected 20 % (i.e., 74) of the 16-min transcript segments and created a table where, for all

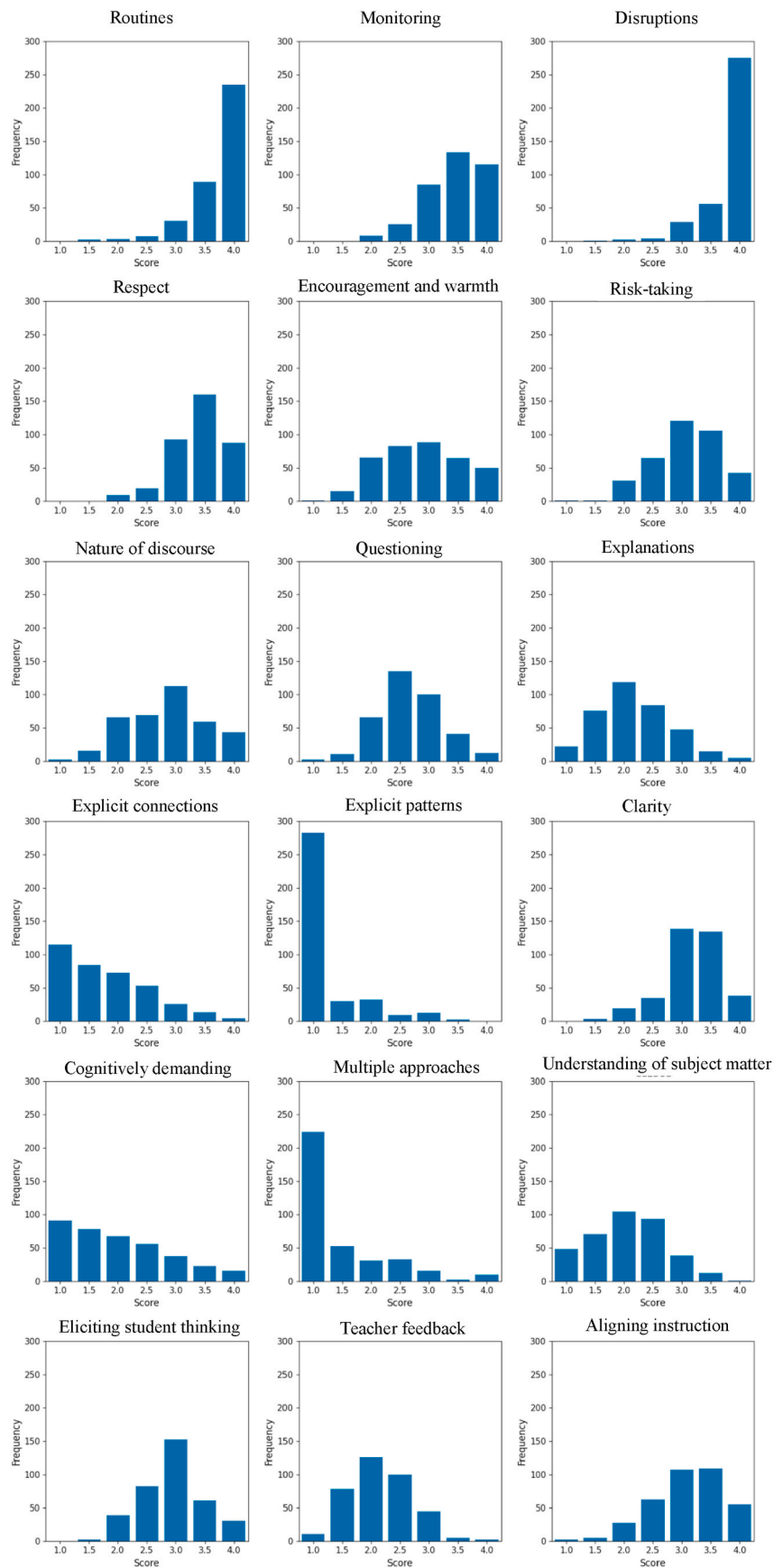


Fig. 2. Distribution of Averaged Human Ratings Across 18 Subdimensions

Note. This figure illustrates the number of instances falling into each score level per subdimension. Due to space limitations, we had to abbreviate the names of some subdimensions.

transcript segments and across all 18 teaching effectiveness subdimensions, the means of the two human-generated scores and the ML-generated scores (i.e., scores from text and audio model⁶) were listed side by side. Next, two human expert raters were tasked with reviewing each transcript segment and each teaching effectiveness dimension where the scores differed. The two raters involved in this plausibility evaluation were trained student assistants who received extensive instruction based on the GTI coding manual. Importantly, these raters were not part of the original pool of human raters who generated the training data, ensuring independence between the model development and validation phases. The task of the two raters was to decide, based on the manual, which score appeared more plausible (i.e., 1,332 decisions in total per human rater [18 decisions per segment]). The expert raters did not know which scores were human-generated or ML-generated. We calculated the percentage of instances where the ML-generated score was deemed more plausible for each dimension of teaching effectiveness. Additionally, we calculated Cohen's kappa as a measure of inter-rater reliability between the two raters.

3.5.5. Predictive validity of ML-generated scores (RQ3)

To gain insights into the predictive validity of the ML-generated scores, we examined whether they predicted student mathematics achievement as well as the human-generated scores did. We performed structural equation modeling (SEM) using the R package *lavaan* version .6.17 (Rosseel, 2012) to investigate the relationships between teaching effectiveness scores and student outcomes. Separate models were estimated for different types of teaching effectiveness scores: (a) human-generated mean scores, (b) ML-generated scores based on text, (c) ML-generated scores based on text and audio, and (d) ML-generated scores based on text, audio, and video. In the first model (M1), the dependent variables were predicted solely by the teaching effectiveness scores. In the second model (M2), we extended M1 by including the teacher-level and student-level covariates (see Table S1 in our OSF project) to account for additional sources of variance. We applied the Benjamini-Hochberg correction (Benjamini & Hochberg, 1995) to control the false discovery rate (i.e., we adjusted the p values for multiple comparisons within each model).

To appropriately model the hierarchical structure of the data—students nested within teachers—we employed cluster-robust standard errors. This approach accounts for the dependency among observations and provides more accurate standard errors without requiring the explicit modeling of random effects (McNeish et al., 2017).

We used Full Information Maximum Likelihood (FIML) estimation to address missing data. FIML is particularly advantageous, as it uses all available data while producing unbiased parameter estimates, assuming data are missing at random. Because FIML outperforms traditional approaches (e.g., listwise deletion), it has been recommended in the literature as a best practice for handling missing data in SEM (Graham, 2012; van Buuren, 2018).

4. Results

4.1. Reliability of ML-generated scores (RQ1)

To gain insight into the reliability of the ML-generated scores for all 18 teaching effectiveness subdimensions, we examined absolute accuracy, relative accuracy, and bias (Table 2). We found some interesting patterns of results.

When looking only at the absolute accuracy, a vertical comparison of the scores in Table 2 (i.e., regarding the different reliability measures)

⁶ We decided to compare the scores of this ML model with the human scores because this model provided encouragement and warmth subdimension scores that were as reliable as human raters' scores in our previous study (Hou et al., 2024).

shows that the ML-generated scores from the text and audio ML models were more reliable than the human-generated scores (i.e., the lowest and, therefore, the most reliable values are present here for 11 subdimensions; in bold in Table 2 and Fig. 3). When examining only the bias, the human-generated scores showed less bias in a specific direction than the ML-generated scores, with the lowest bias values present for 12 subdimensions (in bold in Table 2). Noticeably, as the bias—defined as the sum of the differences between the means of the human-generated scores minus the ML-generated scores—was positive for all three subdimensions of the dimension classroom management (i.e., routines, monitoring, and disruptions), the humans gave higher (i.e., more positive) teaching effectiveness ratings than the ML models did. This pattern suggests ML models may adopt a more conservative scoring approach for observable behaviors, potentially due to training on averaged human ratings that smooth out variability.

For six subdimensions (i.e., routines, monitoring, disruptions, explicit patterns and generalizations, multiple approaches to/perspectives on reasoning, and understanding of subject matter procedures and processes), a horizontal comparison of the scores in Table 2 (i.e., regarding the different teaching effectiveness subdimensions) shows that the human-generated scores had higher reliability than the ML-generated scores for at least two reliability measures. For four subdimensions (i.e., questioning, explanations, teacher feedback, and student-aligned instruction), the scores from the text-based ML model showed higher reliability than the scores from humans and the other ML models for two reliability measures. And for four subdimensions (i.e., nature of discourse, clarity, engagement in a cognitively demanding subject matter, and eliciting student thinking), the scores from the text and audio-based ML model showed higher reliability than the scores from humans and the other ML models for two reliability measures.

For a detailed overview of the correlations between human-generated and ML-generated scores with confidence intervals, means, and standard deviations for each teaching effectiveness subdimension, see Tables S2 to S19 in our OSF project. We also provide a confusion matrix for each subdimension and all ML models.

4.2. Content validity of the ML-generated scores (RQ2)

To gain insight into the content validity of the ML-generated scores, we examined the plausibility ratings of ML-generated scores compared with those of human-generated scores by human expert raters, as displayed in Table 3. To allow for a realistic interpretation, it is important to note that the content validity evaluation was based solely on transcripts, without access to the corresponding video data.

We found that, except for the subdimensions *clarity* and *understanding of subject matter procedures and processes*, for which the inter-rater reliability was just acceptable, the two human expert raters could not agree on whether the human-generated or the ML-generated scores were more plausible (across all 18 subdimensions, the mean kappa was $k_{\text{mean}} = .125$). Interestingly, both raters rated the ML-generated scores as slightly more plausible on average across all 18 subdimensions (Rater 1: 54.16 %, Rater 2: 53.85 %). The human expert raters found the ML-generated scores for the subdimension *engagement in a cognitively demanding subject matter* particularly plausible (Rater 1: 63.64 %, Rater 2: 78.18 %). By contrast, the two subdimensions of classroom management *routines* and *disruption* were the only two in which human expert raters rated the human-generated scores as more plausible.

4.3. Predictive validity of ML-generated scores (RQ3)

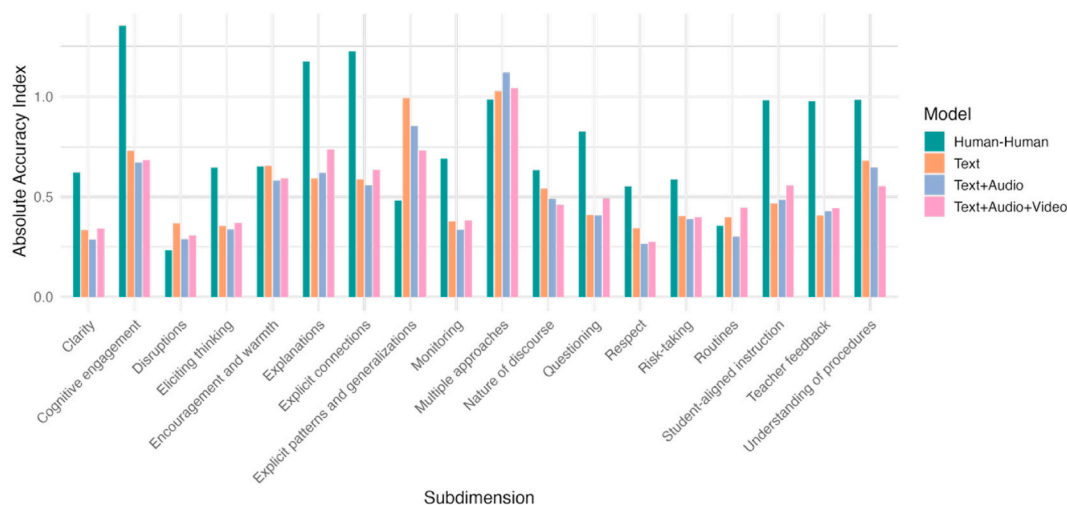
To gain insight into the predictive validity of the ML-generated scores, we examined the estimated parameters (i.e., slopes) of the regressions of students' mathematical achievement on the teaching effectiveness subdimension scores (Table 4). The complete results, including all regression parameters (i.e., including covariates), can be accessed in our OSF project. See Appendix B for an overview of the

Table 2

Reliability of human-generated and machine learning-generated scores of teaching effectiveness subdimensions.

Subdimension	Human-generated scores			ML-generated scores								
				Text			Text + Audio			Text + Audio + Video		
	AA	RA	BS	AA	RA	BS	AA	RA	BS	AA	RA	BS
Routines	.356	.35	.003	.397	.22	.406	.300	.17	.266	.444	.13	.416
Monitoring	.690	.19	.065	.377	.18	.204	.335	.16	.124	.381	.14	.206
Disruptions	.233	.43	.008	.368	.17	.413	.288	.15	.323	.306	.21	.346
Respect	.553	.23	.074	.342	.17	.226	.265	.21	.097	.275	.20	.061
Encouragement and warmth	.652	.51	.050	.654	.26	-.048	.581	.28	-.074	.592	.29	-.048
Risk-taking	.587	.38	.080	.403	.24	-.069	.389	.24	-.065	.397	.20	-.027
Nature of discourse	.634	.51	-.083	.541	.38	.055	.489	.49	-.013	.460	.51	-.062
Questioning	.826	.21	.057	.410	.38	-.005	.407	.37	.053	.493	.20	-.052
Explanations	1.175	.17	.078	.592	.32	-.142	.620	.30	-.161	.737	.21	-.108
Explicit connections	1.226	.29	-.093	.587	.48	-.255	.557	.47	-.260	.634	.39	-.215
Explicit patterns and generalizations	.482	.38	.041	.992	.01	-.692	.852	.02	-.674	.731	.10	-.578
Clarity	.622	.25	-.027	.333	.20	.107	.286	.26	.057	.341	.17	-.027
Engagement in cognitively demanding subject matter	1.354	.38	.080	.730	.47	-.227	.671	.49	-.228	.683	.44	-.212
Multiple approaches to/perspectives on reasoning	.986	.39	.093	1.027	.04	-.561	1.120	.11	-.676	1.042	.03	-.580
Understanding of subject matter procedures and processes	.985	.27	.076	.680	.04	-.101	.646	.03	-.200	.553	.11	-.142
Eliciting student thinking	.646	.30	.045	.354	.34	.096	.336	.38	-.061	.369	.35	.017
Teacher feedback	.978	.11	-.057	.407	.31	-.064	.428	.31	-.072	.443	.25	-.072
Student-aligned instruction	.982	.22	-.023	.467	.33	.037	.485	.22	.066	.556	.16	.091

Note. The metrics for the human-generated scores reflect the agreement between the two human raters. The metrics for the ML-generated scores are based on comparisons with the arithmetic mean of the human-generated scores. AA = absolute accuracy, RA = relative accuracy, BS = bias. The best values per subdimension are marked in bold.

**Fig. 3.** Absolute Accuracy Across Subdimensions and Models

Note. Note that ML scores are continuous and compared to the average of two human ratings, whereas the interrater reliability reflects ordinal scores only. Therefore, ML scores may achieve lower absolute errors partly due to their finer granularity.

means, standard deviations, and correlations of the human-generated scores for the 18 teaching effectiveness subdimensions at the class level. In the following, we focus on the adjusted regression weights in presenting the results.

Regarding the human-generated scores, the subdimensions *questioning* ($b_{adj}^* = .22, p < .001$) and *understanding of subject matter procedures and processes* ($b_{adj}^* = .40, p < .001$) were statistically significant positive predictors of students' mathematics achievement. By contrast, the subdimensions *explicit patterns and generalizations* ($b_{adj}^* = -.24, p = .004$) and *student-aligned instruction* ($b_{adj}^* = -.28, p < .001$) were statistically significant negative predictors.

The ML-generated scores (text-based ML model) showed statistically significantly positive associations between the subdimensions *disruptions* (i.e., less disruption; $b_{adj}^* = .30, p = .025$), *clarity* ($b_{adj}^* = .15, p < .001$), and *eliciting student thinking* ($b_{adj}^* = .28, p = .013$) and students' mathematics achievement. By contrast, the subdimensions *monitoring*

($b_{adj}^* = -.42, p < .001$), *encouragement and warmth* ($b_{adj}^* = -.40, p = .003$), and *multiple approaches to/perspectives on reasoning* ($b_{adj}^* = -.24, p < .001$) were statistically significant negative predictors.

The ML-generated scores (text- and audio-based ML model) showed statistically significantly positive associations between the subdimensions *routines* ($b_{adj}^* = .33, p = .003$) and *engagement in a cognitively demanding subject matter* ($b_{adj}^* = .30, p < .001$) and students' mathematics achievement. By contrast, the subdimensions *encouragement and warmth* ($b_{adj}^* = -.13, p = .009$) and *multiple approaches to/perspectives on reasoning* ($b_{adj}^* = -.47, p < .001$) were statistically significant negative predictors.

For the model scores that incorporated all three modalities (i.e., transcripts, audio, and video), no statistically significant relationships with students' mathematics achievement were found after controlling the false discovery rate.

Table 3

Content validity of the machine-learning-generated teaching effectiveness sub-dimension scores.

Subdimension	<i>n</i>	<i>k</i>	<i>p</i>	% R1	% R2
Routines	52	.025	.812	46.15	15.38
Monitoring	58	.264	.041	53.49	62.07
Disruptions	54	.077	.570	37.04	35.19
Respect	43	.333	.029	44.19	39.53
Encouragement and warmth	64	.281	.024	51.56	50.00
Risk-taking	47	-.123	.391	53.19	61.70
Nature of discourse	54	-.015	.910	57.41	42.59
Questioning	48	-.083	.540	50.00	66.67
Explanations	55	-.103	.440	60.00	52.72
Explicit connections	60	-.033	.795	50.00	55.00
Explicit patterns and generalizations	71	.059	.525	59.15	85.92
Clarity	46	.447	.001	58.70	43.48
Engagement in cognitively demanding subject matter	55	.227	.074	63.64	78.18
Multiple approaches to/perspectives on reasoning	70	.018	.874	65.71	47.14
Understanding of subject matter procedures and processes	49	.413	.003	63.27	55.10
Eliciting student thinking	51	.172	.220	52.94	54.90
Teacher feedback	54	.276	.042	57.41	59.26
Student-aligned instruction	45	.016	.912	51.11	64.44
Mean	54.22	.125	.400	54.16	53.85

Note. Content validity was assessed using plausibility ratings of ML-generated scores in contrast to human-generated scores. The total number (*N*) of evaluated scores was 74.

n = number of scores actually evaluated (i.e., *N* minus number of scores where human and ML-generated scores were identical), *k* = Cohen's kappa, % R1/% R2 = Percentage of *n* scores for which Rater 1 (R1)/Rater 2 (R2) considered the ML-generated scores more plausible.

Table 4

Students' mathematics achievement regressed on the 18 teaching effectiveness subdimension scores.

Subdimension	Human-generated score				Machine-learning model-generated score											
					Text				Text + Audio				Text + Audio + Video			
	M1		M2		M1		M2		M1		M2		M1		M2	
	<i>b</i> [*]	<i>p</i>	<i>b</i> _{adj} [*]	<i>p</i>	<i>b</i> [*]	<i>p</i>	<i>b</i> _{adj} [*]	<i>p</i>	<i>b</i> [*]	<i>p</i>	<i>b</i> _{adj} [*]	<i>p</i>	<i>b</i> [*]	<i>p</i>	<i>b</i> _{adj} [*]	<i>p</i>
Routines	.17	.351	.03	.535	-.33	.228	-.02	.905	.32	.042 ^a	.33	.003	.31	.242	-.05	.672
Monitoring	.09	.236	-.05	.343	.12	.471	-.42	< .001	-.02	.839	-.25	.022 ^a	.26	.058	-.30	.019 ^a
Disruptions	-.17	.129	-.03	.658	.20	.288	.30	.025	-.25	.320	-.16	.265	-.35	.129	.11	.441
Respect	-.06	.162	.04	.487	-.21	.053	-.09	.179	-.15	.242	.17	.049 ^a	-.27	.019 ^a	.08	.214
Encouragement and warmth	-.02	.146	-.02	.719	.14	.325	-.40	.003	.05	.510	-.13	.009	.23	.022 ^a	-.10	.130
Risk-taking	-.08	.039 ^a	-.01	.824	-.11	.427	.22	.029 ^a	.02	.849	.04	.327	-.17	.128	.12	.127
Nature of discourse	-.28	.021 ^a	-.20	.017 ^a	-.11	.604	.08	.474	-.29	.064	-.10	.220	-.44	.005	.23	.044 ^a
Questioning	.23	.428	.22	< .001	.15	.387	.03	.738	.16	.219	-.01	.951	-.30	.004	.21	.014 ^a
Explanations	-.10	.086	-.11	.016 ^a	.14	.280	-.05	.480	.12	.300	-.08	.328	-.42	.002	.09	.418
Explicit connections	.09	.269	-.06	.143	-.12	.246	.01	.867	.12	.240	.11	.235	.15	.127	.07	.307
Explicit patterns and generalizations	-.28	.002	-.24	.004	.21	.053	.02	.751	.09	.528	.15	.025 ^a	.04	.589	.17	.008 ^a
Clarity	-.02	.784	-.04	.306	.21	.070	.15	< .001	.03	.608	.11	.098	-.02	.806	.07	.437
Engagement in cognitively demanding subject matter	-.08	.556	.17	.040 ^a	.12	.385	-.16	.085	.07	.670	.30	< .001	.20	.151	-.15	.123
Multiple approaches to/perspectives on reasoning	-.07	.387	.08	.056	-.26	.028 ^a	-.24	< .001	.17	.053	-.47	< .001	.08	.451	-.25	.009 ^a
Understanding of subject matter procedures and processes	.41	< .001	.40	< .001	.20	.165	.10	.162	-.11	.209	.16	.022 ^a	-.15	.104	-.08	.350
Eliciting student thinking	.14	.218	-.12	.094	-.05	.658	.28	.013	.02	.897	-.01	.910	.01	.948	.08	.547
Teacher feedback	-.07	.358	.06	.300	-.40	.043 ^a	.20	.060	.13	.362	.07	.356	.05	.804	.15	.183
Student-aligned instruction	-.27	.006	-.28	< .001	.35	.085	-.06	.557	-.06	.597	.13	.059	.03	.863	.07	.523

Note. *b*^{*} = unadjusted standardized parameter estimates, *b*_{adj}^{*} = adjusted standardized parameter estimates (i.e., including covariates in the models).

^a Not statistically significant after Benjamini-Hochberg correction.

5. Discussion

The importance of the core dimensions and corresponding sub-dimensions of teaching effectiveness as characteristics of high-quality teaching for successful student learning has been recognized for decades. However, traditional approaches to assessing teaching effectiveness (e.g., student and teacher ratings or classroom observations) face reliability, validity, and scalability challenges in both research and practice. Against this backdrop, in this study, we addressed the limitations of traditional teaching effectiveness assessments by exploring ML algorithms to automate evaluations using multimodal classroom data (video, audio, transcripts) as a scalable and efficient alternative. We compared ML-generated teaching effectiveness scores from real teaching situations with human-generated scores and examined their reliability, content validity, and predictive validity.

5.1. Reliability of ML-generated teaching effectiveness scores

By and large, the ML models were as good as the human raters (even though the quality of the human ratings were less than optimal in some dimensions. However, interesting patterns of findings emerged. It became evident that the reliability of different ML models regarding different combinations of modalities (i.e., text, audio, video) depended on (a) the reliability measure and (b) the teaching effectiveness subdimension.

- (a) On one hand, we found that human-generated scores displayed less bias in 12 of 18 teaching effectiveness subdimensions compared with ML-generated scores. On the other hand, we observed that scores derived from text- and audio-based ML

models demonstrated better absolute accuracy in 11 of 18 teaching effectiveness subdimensions. These findings suggest that human judgments were more balanced overall, avoiding systematic overestimating or underestimating. Higher absolute accuracy of the ML scores suggests that their judgments were numerically closer to the correct values. However, they may still have exhibited systematic bias (e.g., overestimating or underestimating certain subdimensions). Moreover, as ML models trained on text and audio have access to fine-grained linguistic and prosodic features that humans may overlook or inconsistently interpret, these findings could also express that ML can detect exact sentence structure, turn-taking patterns, and word frequencies, contributing to discourse effectiveness. By contrast, human raters may also apply holistic but inconsistent criteria, lowering their absolute accuracy but making them less prone to systematic bias. In general, a high bias should not be interpreted as inherently *bad*; the bias score of an ML score merely indicates whether a teaching effectiveness subdimension is estimated higher or lower than human raters would, whereas its desirability must be evaluated through content and predictive validity.

- (b) We found that the humans showed higher reliability when scoring classroom management subdimensions such as routines, monitoring, or disruptions. It is possible that, like the different classroom management scripts of novice and expert teachers, human raters can apply situational awareness and adapt their judgments to context-specific factors (e.g., teacher's intent, student engagement, and response patterns; see Wolff et al., 2021). ML models rely on predefined patterns and may struggle with ambiguous, context-dependent behaviors that do not fit neatly into their learned structures.

Moreover, we found that subdimensions related to discourse, such as the nature of discourse, questioning, and explanations, could be determined more reliably in our study using text-based or text-and-audio-based ML models than by humans. These subdimensions are defined by characteristics such as the type of questions or teacher- and student-provided written or verbal explanations that are particularly explicit and quantifiable in text-based features, such as sentence structure (e.g., interrogatives for questioning), keyword frequency (e.g., *because* or *therefore* for explanations), and turn-taking patterns (e.g., teacher vs. student discourse balance). Thus, ML models trained on text and audio data may identify linguistic markers better than humans. Humans may overlook subtle patterns or apply inconsistent interpretations, whereas ML models detect patterns systematically.

Finally, we found that subdimensions related to the assessment of and responses to student understanding, such as eliciting student thinking, teacher feedback, and student-aligned instruction, could also be determined more reliably in our study using text-based or text-and-audio-based ML models than by humans. It might be the case that text-based and text-and-audio-based ML models systematically and consistently analyze structured linguistic patterns. By contrast, human raters may differ in how they interpret, weigh, or focus on various aspects of the interaction, resulting in lower interrater reliability. For instance, ML models could detect the frequency and depth of student responses when assessing student thinking by identifying explicit lexical and syntactic features (e.g., sentence complexity and open-ended question structures). Likewise, in evaluating teacher feedback, ML models could reliably monitor specific feedback markers, including clarifications (e.g., "What do you mean by ...?"), affirmations (e.g., "Yes, that's correct because ..."), and error corrections (e.g., "That's incorrect because ..."). Moreover, when analyzing the student-aligned instruction, ML models could evaluate whether teachers effectively incorporate student responses into their instruction by identifying referential connections in dialogue.

Additionally, ML models could track structural and prosodic cues in teacher feedback, making them particularly effective for capturing

complex feedback loops. Whereas text-based ML models detect patterns in textual interactions, audio-based ML models integrate prosodic features such as pauses (potentially indicating reflection time), pitch variations (potentially suggesting emphasis on specific feedback points), and intonation patterns (potentially distinguishing questioning from explaining). By contrast, human raters may subjectively interpret these cues differently, contributing to lower interrater reliability. Moreover, human raters might experience cognitive load, mainly when processing dynamic, multiturn teacher-student interactions. By contrast, ML models may consistently apply predefined decision criteria, leading to higher reliability in evaluating these subdimensions.

In general, the low reliability of human raters (i.e., interrater reliability) stood out in our study. Our results could serve as a renewed impetus to emphasize the criteria for raters, their training (e.g., Praetorius et al., 2012), and the quality of the standardized observation manuals (e.g., Klette & Blikstad-Balas, 2018; Praetorius & Charalambous, 2018). In this regard, ML-generated scores could also serve as a training tool for enhancing rater consistency. Moreover, the truth remains unknown (i.e., the *true* teaching effectiveness score). So far, findings have been based on comparisons of interrater reliability between ML models and human raters, as well as among human raters themselves. Thus, ML models could be further trained on alternative rating sources (e.g., student or teacher ratings) to improve their applicability and robustness.

5.2. Content validity of ML-generated teaching effectiveness scores

In this study, humans trained to assess teaching effectiveness subdimensions based on the GTI manual did not consistently rate the human-generated or ML-generated scores as more plausible. Descriptive analyses even suggested that the ML-generated scores were perceived as slightly more plausible. These results are aligned with findings from studies where participants were asked to distinguish between AI-generated and human-generated content, such as poetry (Porter & Machery, 2024) and student essays (Fleckenstein et al., 2024). These studies have revealed that humans often struggle to differentiate between AI-generated content and content created by humans. We interpret this finding as indicating that ML-generated scores are content-valid, suggesting that ML-generated scores of teaching effectiveness subdimensions may capture essential patterns in teaching effectiveness that are aligned with human intuition.

Interestingly, on the one hand, the human expert raters found the ML-generated scores for the subdimension *engagement in cognitively demanding subject matter* particularly plausible. By contrast, for the two classroom management subdimensions, *routines* and *disruptions*, the human-generated scores were perceived as more plausible. Regarding the subdimension *engagement in cognitively demanding subject matter*, if the ML model integrates text and audio data, it might capture subtle but meaningful engagement patterns that human raters might overlook or assess inconsistently (e.g., speech patterns, turn-taking, and depth of explanations), as human raters may struggle to process this information comprehensively. Moreover, the subdimension is often explicitly reflected in student discourse, teacher questioning, and extended reasoning, which ML models can analyze effectively through transcripts (Demszky et al., 2023). By contrast, human raters may rely on more implicit cues (e.g., classroom atmosphere or teacher behavior) that might introduce less plausibility. Regarding the *routines* and *disruptions* subdimensions, at least two explanations are possible. First, routines and disruptions may encompass rather observable teacher behaviors (Brophy, 1983; Kounin, 1970), making them easier for human raters to assess consistently. These dimensions involve clear, structured actions (e.g., transitions, classroom order), which likely lead to higher agreement among raters, potentially reinforcing the plausibility of human-generated scores. Second, ML models may struggle with contextual nuances (e.g., determining whether disruptions are managed effectively), which could make their assessments seem less intuitive.

Whereas these differentiated plausibility patterns across sub-dimensions are informative, it is important to interpret them in light of the overall low interrater reliability observed in the plausibility study ($\kappa = .125$). This suggests that perceived plausibility may be context- and content-specific. Moreover, this finding highlights the challenges of evaluating content validity through expert judgment across sub-dimensions. Importantly, the two raters were not involved in generating the training labels and brought independent perspectives to the plausibility assessment. However, the plausibility ratings were based on the same rubrics and scoring protocol used for the human annotations in the training phase. Future studies should therefore consider involving a larger pool of raters to enhance the reliability and robustness of content validity assessments.

5.3. Predictive validity of ML-generated teaching effectiveness scores

Regarding predictive validity, the findings were inconsistent for both the human-generated and ML-generated scores. All predictions are estimated while statistically controlling for all other subdimensions of teaching effectiveness, which is critical to keep in mind.

On the one hand, we found positive associations for certain teaching effectiveness subdimensions. The human-generated scores showed positive relationships between questioning and understanding of subject matter procedures and processes and students' mathematics achievement. Similarly, the ML-generated scores (text- and audio-based models) showed positive associations between routines and engagement in cognitively demanding subject matter and student achievement. These results are in line with prior research in suggesting that effective questioning, clear instructional routines, and cognitively engaging tasks are critical for student learning (Klieme et al., 2009; Seidel & Shavelson, 2007).

On the other hand, we found that several subdimensions showed negative associations with student outcomes. Explicit patterns and generalizations and student-aligned instruction (human-generated) negatively predicted student achievement, encouragement, and warmth; and multiple approaches to/perspectives on reasoning (ML-generated) were negative predictors across different models. Reasons for these findings can be discussed from two perspectives: (a) the ML models utilizing multimodal data and (b) research on teaching and learning.

- (a) It is possible that multimodal data integration is crucial for the predictive validity of ML-generated teaching effectiveness scores. Interestingly, after correcting for multiple comparisons, the model incorporating all three modalities (i.e., text, audio, video) did not show any statistically significant relationships with student achievement. This finding suggests that adding more data modalities does not necessarily improve predictive validity and may even introduce noise or leading to redundancy across modalities, making it harder to detect meaningful associations. On the one hand, this finding is surprising as one would expect a better performance when more modalities are included. On the other hand, it is in line with previous research (e.g., Hou et al., 2024). One possibility is that certain instructional subdimensions are best captured through specific modalities (e.g., questioning in text and warmth in audio), whereas multimodal models dilute these effects. These findings highlight the need for more nuanced fusion strategies and careful modality selection in future work.
- (b) Although teacher practices were directly associated with student outcomes in some previous research (e.g., Nilsen & Gustafsson, 2024), it is possible that the assumptions about the relationships between teaching effectiveness dimensions and student outcomes are indeed too simplistic. When conceptualizing teaching as a complex interaction system between students and teachers, these findings are not surprising and are aligned with previous research in showing that direct effects often fail to emerge (Praetorius et al., 2018). For instance, Kleickmann et al. (2020) found that

the teaching effectiveness dimensions did not have the predictive validity that cognitive support had, but, for instance, classroom management and student support were not statistically significantly related to student understanding. The absence of direct effects has frequently been explained by the complexity of teaching effectiveness models (Praetorius et al., 2020), which consider numerous interrelated aspects such as the role of mediating variables (e.g., time on task) and the assumption of opportunity-use structures (Alp Christ et al., 2024; Klieme et al., 2009), making it difficult to generate empirical data within a single study that fully captures relationships. Vieluf and Klieme (2023) went even further and stated: "Even with mediating and moderating factors included, most TER [teaching effectiveness research] still assumes a unidirectional, causal chain connecting teacher behaviour ultimately with student learning. As a consequence, the complexity of the moment-to-moment flow of classroom interactions, with teachers and students sometimes initiating an exchange, sometimes responding to each other (as shown, e.g., by classroom ethnography), as well as the contingencies and ambiguities involved in social interactions are not well understood in the teaching effectiveness paradigm" (p. 63).

Against this backdrop, differential effects may have caused the negative associations found in this study. For instance, teaching strategies that prioritize deep conceptual understanding (e.g., multiple perspectives, explicit generalizations) may initially challenge students, leading to short-term decreases in confidence (self-efficacy) before long-term gains emerge, potentially accompanied by lower student achievement. The subdimension of encouragement and warmth, whereas generally beneficial, could be perceived differently by students in mathematics classrooms—perhaps signaling a compensatory strategy when students struggle rather than an indicator of high achievement. Similarly, frequent monitoring (which was negatively associated with achievement) may signal increased teacher control accompanied by low intrinsic learning motivation or interventions in low-performing classrooms rather than fostering independent learning (e.g., Prenzel, 1997). In addition, knowing that relationships between the provision and use of core teaching effectiveness dimensions have not been found in previous research for these data (Grünkorn et al., 2020) and the fact that ML-generated scores, like human-generated scores, show no clear picture might be interpreted as a positive sign of their validity. These results underscore the need to further intensify the exploration of students' learning processes when researching teaching effectiveness (Seidel & Shavelson, 2007).

Future studies could test mediation models (e.g., via student engagement as per Klieme et al., 2009) to explain these inconsistencies, potentially revealing indirect effects.

5.4. From evidence to impact: where do we stand regarding actionable insights

Our study demonstrates that automated, multimodal assessment of teaching effectiveness is feasible and, in some cases, even matches or exceeds the reliability of human-generated scores. However, to derive actionable insights, it might be valuable to examine the results at the level of individual teaching effectiveness subdimensions. For instance, for the subdimension *engagement in cognitively demanding subject matter*, the text- and audio-based scores show higher absolute and relative accuracy than the human-generated scores, but also exhibit a substantially higher bias, indicating that the automated scores rate this subdimension more positively. Whether this upward bias is problematic cannot be determined solely from the bias metric. Instead, it requires consideration of content and predictive validity. In our case, the automated scores were judged as more plausible (content validity) and statistically significantly predicted students' mathematics achievement when controlling for covariates, whereas the human-generated scores did not.

Such nuanced patterns underscore the importance of careful, subdimension-specific interpretation when discussing actionable insights.

Moreover, the extent to which these ML-generated scores can be translated into actionable insights depends on their application. Whereas the use of ML-generated teaching effectiveness scores in research settings already seems viable, their integration into teacher education and professional development is within reach but still requires refinement. However, the use of such automated assessments in teachers' daily practice remains a distant goal, requiring further advancements in AI technology and research on teaching and learning.

In research contexts, ML-generated scores offer clear advantages, enabling large-scale investigations on teaching effectiveness that were previously limited by the cost and inefficiency of human rating processes. The scalability of ML-generated assessments enables systematic and objective comparisons across classrooms, facilitating studies that would otherwise require extensive observational resources. However, moving beyond research applications and making these assessments meaningful for teacher education and professional development requires additional steps. One promising approach is to provide teachers with detailed, subdimension-level feedback that is enriched with examples of high-quality teaching. If teachers receive not only a numerical score but also contextualized explanations and illustrative examples, ML-generated assessments could serve as a powerful tool for self-reflection and targeted professional growth. The transition from mere scoring to meaningful feedback would make these assessments truly actionable for practitioners.

Despite these opportunities, important challenges remain. One key concern is the reliability of human-generated ratings, which currently served as the ground truth for ML models in our study. Our plausibility ratings revealed substantial disagreement between human raters, potentially underscoring the subjectivity inherent in traditional assessment methods. Thus, the question arises as to whether AI should simply imitate human judgments or move beyond them. If human ratings are used as the benchmark, their quality must be improved—perhaps by aggregating the assessments of multiple highly trained raters to establish a more stable ground truth. Alternatively, future research could explore AI models that learn directly from classroom interactions without relying on human labels. Recent advances in multimodal foundation models, which can process large amounts of text, audio, and video data without extensive labeled training sets, suggest a promising direction for developing more robust and unbiased assessments of teaching effectiveness.

Another critical factor for making ML-generated assessments actionable is explainability. A numerical score alone is of limited use to teachers and researchers unless it is accompanied by an explanation of how it was derived. Future research should explore methods for enhancing the interpretability of AI-generated ratings, for example, by integrating large language models (LLMs) that not only assign scores but also highlight specific moments in a lesson that influenced the assessment. Such an approach would allow teachers and researchers to engage with AI-generated feedback in a more meaningful way, using it as a basis for reflection and improvement.

Moreover, automating the assessment of teaching effectiveness should not be seen as a simple replacement for human raters but as an opportunity to expand the scope and applicability of such evaluations. Whereas ML-generated scores may never be perfect, their scalability allows for entirely new applications, such as providing accessible, individualized feedback for teachers or enabling large-scale research in classrooms without the logistical barriers of human observations. As AI models continue to improve, they have the potential to make teaching effectiveness assessments not only more efficient but also more insightful and impactful. Future research should therefore advance both ML models and teaching and learning research in tandem to ensure that automated assessments are not only technically feasible but also practically meaningful.

Ultimately, as AI-based systems for evaluating teaching effectiveness become increasingly scalable and integrated into educational contexts, it is critical to consider the associated ethical risks. Although the GTI dataset used in this study was de-identified and collected under strict ethical protocols, future applications may involve real-time, longitudinal monitoring of student behavior. If such data are leaked or used beyond their intended educational purposes—e.g., by universities or employers—this could have severe implications for student autonomy and equity. In a recent systematic review on AI in classrooms (Fütterer et al., 2025), we found that ethical considerations are often given limited attention in empirical studies. This highlights an urgent need for clear, normative guidelines, transparent data governance, and robust privacy protections to ensure that AI-based tools support, rather than undermine, educational fairness. Importantly, such technologies could always be used to assist and empower them in meaningful, developmentally appropriate ways, but also to control students' learning or teachers' instructional practices.

5.5. Strengths, weaknesses, and future research

Our study represents an important step forward in the automated assessment of teaching effectiveness dimensions by leveraging multimodal data sources. This study is the first to comprehensively predict teaching effectiveness dimensions using multimodal data, including text, audio, and video. Previous studies have often relied on single-modal approaches, limiting the scope of automated assessment. Our findings provide empirical evidence for the feasibility of multimodal ML models in capturing complex classroom dynamics. Additionally, the comparative evaluation of different ML pipelines against human-generated scores highlights the potential of AI-driven approaches to improve traditional methods of assessing teaching effectiveness.

However, whereas the findings are promising, there are several limitations that should be addressed in future research to further enhance the reliability, validity, and scalability of ML-based teaching assessment models.

First, the content validity evaluation was based solely on transcripts, without access to video data. Thus, raters' abilities to fully assess the plausibility of ML-generated scores may have been restricted, as nonverbal cues (e.g., gestures, facial expressions) were not considered. Future research should incorporate video-based plausibility assessments to provide a more holistic validation of automated teaching assessments.

Second, we employed a specific ML pipeline that has yielded strong results in prior studies. Current ML models replicate human judgments, but AI could be leveraged to go beyond human ratings and generate novel assessments that identify complex classroom interactions (Vieluf & Klieme, 2023). However, alternative ML approaches, such as LLMs, could also be explored. Future studies should experiment with different ML architectures, particularly unsupervised and foundation models, which could improve performance by learning complex interactions without relying on human-labeled training data. This approach could improve validity by enabling ML models to capture latent teaching effectiveness patterns, even if the exact decision-making process remains opaque (black-box issue).

Third, the ML models were trained on human-generated scores, which are known for not always demonstrating high reliability as also shown in this study. This low reliability may limit model performance, as the models are learning from ratings with inherent variability. A critical next step is to develop multimodal foundation models that can learn unsupervised representations of teaching effectiveness, reducing dependence on human labels. Moreover, the AI rater was trained on the mean of only two human raters, which might not be the most robust ground truth. Future studies should aggregate scores from a larger pool of raters (e.g., 10 human raters) to reduce variability. Moreover, alternative ground truths should be explored, such as aggregating classroom-level student ratings or teacher self-assessments to capture multiple perspectives on teaching effectiveness.

Fourth, we relied on pre-existing secondary data, which constrained data quality (e.g., not all students were visible in the videos, blackboard artifacts were partially captured, background noise was present in audio recordings). Future research should work with higher resolution classroom recordings and better data collection methods to mitigate these issues. Additionally, as more powerful AI models become available, they can better process higher quality multimodal data to enhance predictive power.

Further, whereas this study employed teacher-independent cross-validation to evaluate model generalizability to individual teachers, it did not assess potential biases across teacher subgroups, such as their demographic characteristics (e.g., gender or years of experience). This is a critical consideration, particularly when the nature of teaching effectiveness evaluations is of high stakes. Future research should systematically examine model performance across teacher demographics and explore multiple bias mitigation approaches, including post-processing calibration methods and ensemble techniques that combine predictions from models trained on demographically balanced subsets. Additionally, a human-in-the-loop approach, which involves diverse stakeholders such as teachers and administrators in the bias assessment process, could help ensure that mitigation strategies are aligned with real-world concerns.

Finally, the ML-generated scores did not demonstrate predictive validity, but this finding was not entirely unexpected given the data set (Grünkorn et al., 2020). Future research should further explore predictive validity using different data sets where stronger associations between teaching effectiveness and student outcomes have previously been established. Additionally, the generalizability of the findings should be tested across new data sets and diverse educational contexts.

In conclusion, this study provides a strong foundation for the automated assessment of teaching effectiveness dimensions using multimodal AI models. Whereas the findings highlight significant progress,

they also underscore the need for continued refinement, validation, and methodological advancements to fully unlock the potential of AI in classroom teaching analysis, something we aim to address in the newly established Emerging Field Group of the European Association for Research on Learning and Instruction (<https://www.earli.org/efg-13>).

CRediT authorship contribution statement

Tim Fütterer: Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Ruikun Hou:** Writing – original draft, Formal analysis. **Babette Bühler:** Writing – review & editing. **Efe Bozkir:** Writing – review & editing. **Courtney Bell:** Writing – review & editing. **Enkelejda Kasneci:** Writing – review & editing. **Peter Gerjets:** Writing – review & editing. **Ulrich Trautwein:** Writing – review & editing, Conceptualization.

Declaration of competing interests

We have no known conflicts of interest to disclose. Supplementary Material of the study can be accessed at our Open Science Framework (OSF) project: <https://doi.org/10.17605/OSF.IO/9GXXMH>. The authors are responsible for the contents of this publication.

We used Grammarly (1.98.2.0) and ChatGPT-4o to perform language editing to the manuscript.

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Appendix A

Formula of Reliability Scores Calculated for Each Teaching Effectiveness Subdimension.

1)

$$\text{Absolute Accuracy Index} = \frac{1}{N} \sum_{i=1}^N (c_i - p_i)^2$$

2)

Relative Accuracy Index = Pearson's r

3)

$$\text{Bias Index} = \frac{1}{N} \sum_{i=1}^N (c_i - p_i)$$

with N = total number of teaching units for which scores are provided per teaching effectiveness subdimension, c_i human-generated score of a teaching effectiveness subdimension for teaching unit i , and p_i = ML-generated of a teaching effectiveness subdimension for teaching unit i .

Appendix B. Means, Standard Deviations, and Correlations of the Human-Generated Scores for the 18 Teaching Effectiveness Subdimensions at the Class Level

Variable	M	SD	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
1. Routines	3.75	.24																	
2. Monitoring	3.46	.30	.36*																
3. Disruptions	3.83	.18	.65**	.51**															
4. Respect	3.43	.29	.46**	.28	.68**														
5. Encouragement and warmth	2.86	.51	-.16	-.05	-.11	.23													
6. Risk-taking	3.09	.32	-.07	-.14	-.13	.09	.41**												
7. Nature of discourse	2.82	.45	-.06	.15	.04	.10	.18	.18											
8. Questioning	2.66	.36	.03	.23	.12	.03	.24	.13	.53**										
9. Explanations	2.17	.33	-.20	-.13	-.20	.03	.01	.24	-.09	-.23									
10. Explicit connections	1.78	.38	.09	.01	.31*	.12	-.14	-.02	-.04	.12	-.10								
11. Explicit patterns and generalizations	1.25	.31	-.23	.20	-.11	-.31*	-.11	.16	.08	.32*	-.11	-.13							
12. Clarity	3.19	.26	.37*	.21	.42**	.42**	.02	-.16	-.26	-.02	-.19	.07	-.12						
13. Engagement in cognitively demanding subject matter	1.99	.51	.03	.19	.15	.04	.16	.07	.49**	.73**	-.09	.41**	.21	-.11					
14. Multiple approaches to/perspectives on reasoning	1.47	.48	-.01	.10	.05	-.12	-.24	.04	.31*	.29	.03	.00	.39**	-.15	.35*				
15. Understanding of subject matter procedures and processes	2.06	.37	-.03	.15	.08	.13	.17	.34*	.43**	.28	.41**	.07	.33*	-.07	.38*	.43**			
16. Eliciting student thinking	2.94	.32	.06	.22	.18	.18	.41**	.10	.81**	.72**	-.15	.19	.13	-.06	.72**	.28	.41**		
17. Teacher feedback	2.16	.26	.05	-.10	.01	.04	.07	.30*	.26	.04	.50**	-.07	-.15	-.35*	.06	-.00	.31*	.12	
18. Student-aligned instruction	3.11	.34	.16	.28	.08	.06	.16	.10	.11	.38*	.18	.07	.08	-.05	.45**	.11	.42**	.23	.35*

Note. M and SD are used to represent mean and standard deviation, respectively.

* $p < .05$. ** $p < .01$.

Data availability

The data that has been used is confidential.

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